

GOVERNING PRINCIPLES FOR ALL DATA SOURCES	1. FREE - DIRECT FROM PRIMARY SOURCE Prefer official free API or CSV endpoints from the data creator (FRED, CBOE, CFTC, BLS). These are tier-1 — no third-party dependency, no subscription risk. 2. YAHOO FINANCE Currently the main dependency for all market prices (via yfinance / load_yf — confirmed throughout codebase). Acknowledged single point of failure. Address in a future phase. 3. SCRAPING Acceptable ONLY where no API exists (AAII, NAAIM, multpl.com). Must have retry logic + cached fallback. AAIL: confirmed in codebase — Ahil built a direct .xls download from aail.com/files/surveys/sentiment.xls via requests.get() with browser headers. This is a download, not a page scrape — more stable. 4. NO SUBSCRIPTIONS No variable in this system may depend on a paid personal subscription (StockCharts, mcoscillator.com) that could lapse. Compute from primary components instead. 5. GOVERNMENT SHUTDOWN RISK Applies to BLS (CPI) and CFTC. Both have been interrupted by shutdowns. Apply identical retry/cache logic: retry same day, retry next day, preserve last known SQLite value — never write NULL.
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CODEBASE FINDINGS (from uploaded files)	1. ALL PRICE DATA via load_yf() / Yahoo Finance (yfinance) — confirmed across sentiment.py, compute.py, fib_ret.py, bollinger.py, distance.py, new_high.py, divergence_comp.py. This is the universal price source. 2. 200-DAY MA already computed in codebase — fib_ret.py:add_moving_averages() computes df['Close'].rolling(200).mean() per symbol using daily data. fib_ext.py and fract_track_bands.py also compute ma_200d inline. CONCLUSION: %stocks above 200DMA should be computed directly using existing MA200 infrastructure across S&P; 500 constituents — do NOT add an external dependency on S5TH or ^SPXA200R. 3. AAIL DATA — confirmed in util.py:download_aail_sentiment_file(). Uses requests.get() on https://www.aail.com/files/surveys/sentiment.xls with browser headers. Saves to sentiment.xls. Read back in sentiment.py via pd.read_excel('/sentiment.xls'). This is already built — do not rebuild. Divyanshu: reuse this function for the SSI layer. 4. SPREAD / BULL / BEAR from AAIL used directly in sentiment.py trade logic: BUY_THRESHOLD=10 (spread>10), SELL via inverse-U shape detection on spread. RSI confirms direction. 5. NO FRED, CBOE, CFTC, BLS, CNN F&G;, NAAIM, McClellan, SKEW, DBMF, HYG/LQD pulls found in codebase. All macro/SSI variables for the new Runic Agent and full SSI build are NEW additions — not duplicating existing code.
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COMPLETE VERIFIED DATA SOURCE TABLE — 26 VARIABLES

#	Variable	System / Layer	Release Frequency	Release Day & Time ET	Source / Ticker	Pull Method (API? Free?)	Lookback Window	≥2 Sources Confirmed	Notes (authoritative)
MACRO INTELLIGENCE AGENT — 12 VARIABLES									
1	NFCI (Chicago Fed Financial Conditions)	Macro Runic	Weekly	Wed ~8:30am ET	FRED: NFCI	fredapi Python pkg ■ Free. Stable 15+ years.	Expanding window from 1973 (official series start)	■ FRED + Chicago Fed officer site	NFCI constructed to mean=0, SD=1 since 1973 by Chicago Fed. Positive=tighter than avg, negative=looser. Current: -0.523 = LOOSE. Not at RARE threshold (+0.3). Same data served by fredapi and directly from Chicago Fed. Updated Wednesday covering prior Friday data.
2	HY Credit Spreads OAS	Macro Runic	Daily	EOD — FRED updates next morning	FRED: BAMLH0A0HYM2	fredapi Python pkg ■ Free. ICE BofA series.	Expanding window from 1996 (FRED series start)	■ FRED + GuruFocus (274bps current, May 2026)	Full historical range 2.41%–21.82%. Long-term avg 3.94%. Current ~305bps = NORMAL, ~20th pctile. SSI Layer 2 uses HYG/LQD ratio (same underlying). RULE: do NOT double-count — FRED OAS feeds Runic; HYG/LQD feeds SSI. One pull each, separate signals.
3	Fed Balance Sheet MoM Δ% (WALCL)	Macro Runic	Weekly	Thu ~4:30pm ET (H.4.1 release)	FRED: WALCL	fredapi Python pkg ■ Free.	Expanding from 2008 (QE era start — pre-QE regime excluded)	■ FRED H.4.1 release schedule confirms Thu	MoM % change = scale-invariant. Pre-2008 data structurally different (balance sheet fraction of today's). Same shutdown risk as BLS/CFTC — apply retry/cache logic. Store in SQLite; update weekly.
4	USD/CNH 4-week % change	Macro Runic	Daily (compute weekly)	Continuous. Compute EOD Friday	Yahoo Finance: USDCNH=X	yfinance (load_yf) ■ Free. In codebase.	Expanding from 2010 (CNH offshore market established 2010)	■ Yahoo Finance + Investing.com USDCNH	CNH freely floated only post-2010. Pre-2010 data not comparable. Compute 4-week rolling % change every Friday. Store in SQLite.
5	WTI Oil 4-week % change	Macro Runic	Daily (compute weekly)	Continuous. Compute EOD Friday	Yahoo Finance: CL=F	yfinance (load_yf) ■ Free. In codebase.	Expanding from 1985 (NYMEX WTI futures launch)	■ Yahoo Finance + TradingEconomics (\$87.59 confirmed May 2026)	Use 4-week % change (scale-invariant — no inflation adjustment needed for rate-of-change). For absolute level comparisons across decades: use WTI/CPI deflated. Current: ~-17% over 4 weeks (falling from \$117 peak). Combo C CANCELLATION APPROACHING.
6	VIX Level	Macro Runic	Daily	EOD ~4:15pm ET	Yahoo Finance: ^VIX (CBOE primary source)	yfinance ■ Free. CBOE cboe.com also confirms.	Expanding from 1990 (CBOE VIX inception — cannot extend further)	■ Yahoo Finance ^VIX + CBOE cboe.com/tradable-products/vix (15.32 confirmed May 29, 2026)	DUAL paradigm: >25 AND >80th pctile. The dual condition prevents false triggers during sustained low-vol regimes. VIX regime multiplier (SSI Layer 4) is BYPASSED when Combo B fires — critical rule preventing the Dalio Oct 2022 error. Runic use only — excluded from SSI to avoid double-counting.
7	VIX Term Structure (VIX3M ÷ VIX)	Macro Runic + SSI Layer 2	Daily	EOD ~4:15pm ET	Yahoo Finance: ^VIX3M ÷ ^VIX	yfinance ■ Free. ^VIX3M confirmed active May 28, 2026 (19.11).	Expanding from 2007 (VIX3M inception — cannot extend further)	■ Yahoo Finance ^VIX3M + Google Finance VIX3M:INDEXCBOE	Normal long-run avg ratio = 1.07–1.10. Ratio >1.10 = near-term complacency above average. Today ~1.19 = WATCH (Combo D thresh 1.10). Backwardation (<1.0) = panic peak = Combo G + capitulation signal. NOTE: same calculation used in both Runic (Combo D/G) and SSI Layer 2. One pull, used twice. Do NOT double-count in combined SSI+Runic score.
8	CFTC Net Spec Positioning (Fast Money / Leveraged Funds)	Macro Runic + SSI Layer 3	Weekly	Fri 3:30pm ET (covers Tue data — 3-day inherent lag)	CFTC.gov TFF report direct CSV download	requests.get() on CFTC CSV ■ Free. Direct from regulator. One download serves vars 8,23,24,25.	3-year rolling window (156 weeks) — industry standard for COT positioning	■ CFTC.gov official page + TradeAlgo / ForexFundamentals (3yr rolling = documented standard)	■■ GOVERNMENT SHUTDOWN RISK: CFTC data interrupted Oct 1–Nov 12, 2025. Apply same retry/cache logic as BLS. Use FAST MONEY (Leveraged Funds = hedge funds/CTAs) as primary combo signal. Real Money (Asset Mgr) stored separately for SSI Squeeze Setup/Liquidity Exit. Extreme: >90th or <10th pctile per industry standard.
9	10Y–2Y Yield Curve	Macro Runic	Daily	~4:15pm ET (Fed H.15 release)	FRED: T10Y2Y	fredapi Python pkg ■ Free.	Expanding from 1976 (FRED series start; 12,000+ daily obs, no revisions ever)	■ FRED T10Y2Y + Eco3min.fr yield curve dataset (50 years, confirmed active)	Every inversion since 1976 preceded a US recession. DUAL paradigm: level AND rate of steepening. The steepening RATE post-inversion is the key timing signal — not the inversion level itself. Current: +43bps, steepening = post-inversion steepening phase active.
10	CPI Surprise vs Consensus	Macro Runic	Monthly	~2nd Tue/Wed of month 8:30am ET (BLS)	BLS API: CUSR0000SA0 (primary) FRED: CPIAUCSL (fallback — 2+ day lag)	BLS API: requests.post() ■ Free. Retry queue. FRED fallback if BLS unavailable 2+ consecutive days only.	3-year rolling window (consensus accuracy changes over time)	■ BLS.gov official API + FRED CPIAUCSL series (confirmed active)	■■ GOVERNMENT SHUTDOWN RISK: Oct 2025 CPI CANCELLED ENTIRELY due to shutdown. Apply retry queue: attempt BLS at 8:30, 9:00, 10:00, 13:00, 16:00, 20:00 ET on release day. ONLY fall back to FRED if BLS unavailable for 2+ full calendar days AND FRED has confirmed the new print (verify via FRED release calendar API). NEVER default CPI to 'not hot' during unavailability — keep last known state. Cache release schedule from bls.gov/schedule/news_release/cpi.htm at startup.

#	Variable	System / Layer	Release Frequency	Release Day & Time ET	Source / Ticker	Pull Method (API? Free?)	Lookback Window	≥2 Sources Confirmed	Notes (authoritative)
11	Gold/Silver Ratio (GSR) 4-week % change	Macro Runic	Daily (compute weekly)	Continuous. Compute EOD Friday	Yahoo Finance: GC=F + SI=F	yfinance ■ Free. Both futures confirmed active.	Expanding from 1968 (consistent modern precious metals data)	■ Yahoo Finance GC=F + SI=F + goldsilver.com (GSR history confirmed)	Use COMEX futures GC=F and SI=F — NOT goldsilver.com (no API, retail site). Compute ratio daily. Compute 4-week % change every Friday. GSR used directionally — rising GSR = risk-off / defensive. Absolute GSR level >80 historically = silver cheap relative to gold. NFTRH (Gary Tanashian) uses directional approach — 4-week % change is the correct signal metric.
12	CAPE / Shiller P/E	Macro Runic	DAILY (updates with market price)	Intraday — reflects current price vs 10yr real earnings	multpl.com (primary) GuruFocus (backup)	HTML scrape ■■■ No API. multpl.com confirmed active (42.66 today May 2026). Backup: GuruFocus API (free tier).	Expanding from 1881 (Shiller's official series — all institutional users use full history)	■ multpl.com (42.66) + GuruFocus (41.33 — minor methodology diff) + Shiller's own data at shillerdata.com	■■■ CORRECTION FROM PRIOR TABLE: multpl.com updates DAILY with current market price, not monthly. Pull weekly on Fridays. Current 42.04–42.66 = EXTREME (>32 threshold) = 2nd highest in 145-year history after dot-com peak of 44.2 in Jan 2000 = ~98–99th percentile since 1881. Minor discrepancy between sources (methodology) — use multpl.com as primary (officially publishes Shiller's data).
SENTIMENT SUPERINDEX — LAYER 1: WEEKLY PULSE (4 signals)									
13	AAII Bull-Bear Spread	SSI Layer 1	Weekly	Thu ~9am ET	aaai.com/files/surveys/sentiment.xls (direct .xls download)	requests.get() + browser headers ■ Free. ALREADY IN CODEBASE. util.py:download_aaai_sentiment_file()	N/A — raw survey data. Use full available history from 1987	■ AAII.com official + MacroMicro.me (publishes Thu ET — confirmed)	■■■ ALREADY BUILT: Ahil implemented download_aaai_sentiment_file() in util.py — direct .xls download from aaai.com/files/surveys/sentiment.xls. sentiment.py reads it via pd.read_excel('./sentiment.xls'). Divyanshu: REUSE this function, do not rebuild. TrendPulse/PulseGauge already uses AAII Spread + Bullish + Bearish % in trade signals (BUY_THRESHOLD=10 on spread, RSI confirmation). Add retry logic and cache for robustness.
14	NAAIM Exposure Index	SSI Layer 1	Weekly	Thu during market hours (survey collected Wed EOD)	naaim.org scrape	HTML scrape ■■■ No API. Retry logic required. Holiday may shift to Friday.	N/A — raw survey data. Full history from 2006 (inception)	■ naaim.org official + MacroMicro (weekly, Thu) + Proactive Advisor Magazine (Thursday release confirmed)	Surveys professional active money managers on % equity exposure (range ~200% to +200%). Reading <30 = extreme defensive = potential buy setup. NOT a contrarian signal at high readings (unlike AAII) — high readings suggest momentum that can persist. scrape naaim.org weekly after market hours Thursday.
15	CBOE Put/Call Ratio (10-week EMA)	SSI Layer 1	Daily (compute 10wk EMA weekly)	EOD after close (CBOE updates same day)	CBOE direct CSV: cdn.cboe.com/resources/options/volume_and_call_put_ratios/equitypc.csv	Single requests.get() call on CBOE's own CDN ■ Free. No API key. No scraping. Direct from primary source.	Expanding from 2006 (CBOE equity P/C series inception)	■ cdn.cboe.com CSV confirmed + cboe.com/us/options/market_statistics/ daily/ confirmed active Apr 2026	The equity put/call CSV lives directly on CBOE's own CDN at cdn.cboe.com/resources/options/.../equitypc.csv. Fetched with a single requests.get() call — no API key, no authentication, no scraping of HTML. Direct from the data creator. Tier-1 source. Compute 10-week EMA from daily history stored in SQLite. Previous suggestion of Yahoo Finance ^PCCE is INCORRECT — use CBOE direct only.
16	CNN Fear & Greed Index	SSI Layer 1	Daily (intraday updates)	Continuous intraday updates	CNN API endpoint: production.dataviz.cnn.io/index/fearandgreed/graphdata/ OR pip install fear-and-greed	HTTP GET to CNN CDN endpoint ■ Free. PyPI package available. fear-and-greed package on PyPI.	Expanding from 2012 (CNN data available from 2012 via archive)	■ CNN production endpoint (confirmed active, GitHub whit3rabbitt/fear-greed-data auto-refreshes every Friday)	Live endpoint: https://production.dataviz.cnn.io/index/fearandgreed/graphdata/ — returns JSON with current score and 7 sub-components. PyPI package 'fear-and-greed' wraps this cleanly. Scale 0–100: <20=extreme fear, >80=extreme greed. Pull daily for SSI. Historical data available from 2012.
SENTIMENT SUPERINDEX — LAYER 2: DAILY TIMING (6 signals)									
17	VIX Term Structure (SSI Layer 2 instance)	SSI Layer 2	Daily	EOD ~4:15pm ET	Same data as var #7 (^VIX3M ÷ ^VIX)	Same yfinance pull as #7. One pull, used in both Runic and SSI.	Same as var #7: expanding from 2007	Same as var #7 (two confirmed sources)	SAME CALCULATION as Runic variable #7. One yfinance pull on Friday, result stored in SQLite, read by both Runic (combo D/G) and SSI (Layer 2 signal). Do NOT pull twice. Do NOT double-count in combined SSI+Runic confidence score — this is the same underlying data point.
18	% S&P; 500 Stocks Above 200-Day MA	SSI Layer 2	Daily	EOD (all stocks close by 4pm ET)	COMPUTE DIRECTLY from existing codebase + S&P; 500 constituent list	Existing MA200 code in fib_ret.py:add_moving_averages() + constituent list from financialmodellingprep.com (free /sp500_constituent endpoint).	S5TH confirmed = S&P; 500 only (not NYSE). Compute fresh daily.	■ Investing.com S5TH confirms = S&P; 500 specifically ■ fib_ret.py codebase has MA200 already implemented	■■■ KEY CODEBASE FINDING: 200DMA already computed per symbol in fib_ret.py:add_moving_averages() using df['Close'].rolling(200).mean(). COMPUTE THIS DIRECTLY: pull S&P; 500 constituent list (503 stocks, updates quarterly), check price>MA200 for each, compute percentage. Do NOT add external dependency on S5TH or ^SPXA200R — Yahoo Finance ^SPXA200R NOT confirmed active. Refresh constituent list after each quarterly S&P; rebalance.
19	McClellan Oscillator (NYSE breadth momentum)	SSI Layer 2	Daily	EOD	COMPUTE DIRECTLY from NYSE daily advance/decline data. A/D from Alpha Vantage free tier or Barchart API.	Alpha Vantage free API (daily A/D data). Formula: 19d EMA(NetAdv) – 39d EMA(NetAdv) where NetAdv = Adv – Decl.	Expanding from calculation start date (compute from stored A/D)	■ StockCharts \$NYMO (confirmed active) ■ mco scillator.com (creators — confirmed Mar 2026)	■■■ DO NOT use mcoscillator.com for production — paid subscription only, no free API. DO NOT rely on personal StockCharts membership — subscription lapse risk. Compute directly from NYSE advance/decline data. Store daily A/D in SQLite. Recompute McClellan EMA each day. Signal: below –80 = oversold breadth = strong buy setup. Below –100 historically marks major market bottoms.

#	Variable	System / Layer	Release Frequency	Release Day & Time ET	Source / Ticker	Pull Method (API? Free?)	Lookback Window	≥2 Sources Confirmed	Notes (authoritative)
20	New Highs / New Lows Ratio (NYSE)	SSI Layer 2	Daily	EOD	StockCharts \$NYHGH and \$NYLOW (primary) OR compute from Barchart API (free tier)	StockCharts free web access for daily data. Alternative: Barchart free API.	Expanding from calculation start date	■■ Yahoo Finance tickers (^NAHGH/^NALOW) NOT confirmed active. Divyanshu MUST test first.	■■ TEST REQUIRED: Yahoo Finance tickers for NYSE NH/NL are unconfirmed. Divyanshu must verify ^NAHGH and ^NALOW on Yahoo Finance before using. If inactive, use StockCharts \$NYHGH/\$NYLOW or Barchart free API. Signal: NH/NL ratio near 0 = broad market washout = buy setup. Positive divergence (ratio recovering while index still falling) is most reliable early bottom signal.
21	HYG/LQD Ratio (HY vs IG credit)	SSI Layer 2	Daily	EOD ~4pm ET	Yahoo Finance: HYG ÷ LQD (both ETF prices)	yfinance ■ Free. Both ETFs active and high-liquidity.	Expanding from 2007 (HYG inception date)	■ Yahoo Finance HYG ■ Yahoo Finance LQD (both confirmed active high-volume ETFs)	SSI uses HYG/LQD price ratio. Runic uses FRED OAS spread. SAME UNDERLYING CREDIT STRESS — do NOT double-count in combined score. HYG/LQD 4-week % change falling = HY underperforming IG = credit stress building = Combo G warning. When both HYG/LQD AND VIX TS diverge (credit widening but VIX flat) = textbook Combo G hidden stress setup.
22	CBOE SKEW Index	SSI Layer 2	Daily	EOD ~5pm ET	Yahoo Finance: ^SKEW	yfinance ■ Free. ^SKEW confirmed 144.18 May 29, 2026.	Expanding from 1990 (SKEW inception date)	■ Yahoo Finance ^SKEW (May 29, 2026 data confirmed) ■ CBOE SKEW dashboard cboe.com/us/indices/dashboard/skew	CONFIRMATION SIGNAL ONLY — not standalone. High SKEW = informed money buying crash protection (OTM puts). Amplifies conviction of other bearish signals. Used as a secondary filter, not a primary trigger. Normal range roughly 100–150; above 150 = elevated tail-risk hedging.
SENTIMENT SUPERINDEX — LAYER 3: POSITIONING (4 signals)									
23	CFTC Fast Money (Leveraged Funds = hedge funds / CTAs)	SSI Layer 3	Weekly	Fri 3:30pm ET (same file as var #8)	CFTC.gov TFF report: Lev_Money_Positions_Long_All and Short_All	Same single requests.get() CSV download as var #8. ■ Free. Direct from CFTC.	3-year rolling window (same as var #8 — documented institutional standard for COT)	Same as var #8 (two confirmed sources)	PRIMARY COMBO SIGNAL for Combo B/D/E/F thresholds. Fast Money = Leveraged Funds in TFF column. <15th pctile = crowded short = Combo B condition. >85th pctile = crowded long = Combo D condition. ■■ SAME FILE as var #8 — download once, parse both FM and RM columns. Same government shutdown risk applies.
24	CFTC Real Money (Asset Managers = pension funds / SWFs)	SSI Layer 3	Weekly	Fri 3:30pm ET (same file as var #8)	CFTC.gov TFF report: Asset_Mgr_Positions_Long_All and Short_All	Same single download as vars #8 and #23. Parse Asset_Mgr columns.	3-year rolling window (same as var #8)	Same as var #8	Used for SSI Squeeze Setup (FM<30th AND RM>50th) and Liquidity Exit (RM<30th AND FM>60th). NOT used for Runic combo B/D/E/F thresholds — those use FM only. RM stored separately in SQLite. Same shutdown risk. One download serves vars 8, 23, 24, 25.
25	Gross/Net Divergence (derived from #23 + #24)	SSI Layer 3	Weekly	Derived — computes after Fri CFTC pull	Computed from vars #23 and #24	Derived calculation. No additional data source.	N/A — derived	N/A — derived	Computed from CFTC FM and RM data already pulled. Gross exposure = FM_Long + FM_Short + RM_Long + RM_Short. Net = (FM_Long – FM_Short) + (RM_Long – RM_Short). Signal fires when gross >75th pctile 3yr rolling for 3+ consecutive weeks AND net falling. Requires concurrent HYG/LQD widening for confirmation — reduces false positives from tactical hedging.
26	DBMF (CTA proxy via RBSA)	SSI Layer 3	Daily	EOD ~4pm ET	Yahoo Finance: DBMF + SPY	yfinance ■ Free. DBMF confirmed \$31.00 May 29, 2026 (AUM \$3.54B).	21-day rolling beta (recomputed daily from stored history)	■ Yahoo Finance DBMF (active, high volume) ■ iMGP/DBi official site ■ Morningstar (AUM \$3.54B)	Compute 21-day rolling beta of DBMF vs SPY using OLS regression. Beta < –0.10 = CTAs likely net short equities. Beta < –0.20 with t-stat > 2.0 = confirmed CTA short with conviction. Provides real-time CTA positioning estimate between Friday CFTC reports (3-day lag offset). World's largest managed futures ETF — tracks top 20 CTAs. Store daily DBMF and SPY prices in SQLite; recompute beta on each new day.